Underspecification prevents linguistic structure in cultural evolution

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Human language refers to the objective external world while encoding subjective internal meanings shaped through speaker's conceptualization. Consequently, it is fundamentally underspecified, exhibiting an asymmetry between a rich internal meaning space (M) and the limited linguistic signals (S). This study modifies the Neural Iterated Learning Model to test the emergence of compositionality under underspecification ($M > S$) with a hearer's bias toward speaker's conceptualization. We implement this and a "conceptualization bit". While the symmetric baseline ($M = S$) successfully yields high–compositionality, compositional structure does not reliably emerge in $M > S$. These results highlight a key assumption in prior ILM work—implicit signal–meaning symmetry—which abstracts away from underspecification. Our findings suggest that resolving underspecification may require mechanisms beyond vertical transmission alone, potentially including pragmatic inference that shapes how meanings become syntactically encoded. This supports the view that pragmatic communication is central to establishing the syntax–pragmatics boundary in language evolution.

1. Introduction

Human communication is not merely the transmission of information. Human language not only refers to objective situations in the external world but also encodes subjective meanings through conceptualization, the process by which language users construe the world from a particular perspective (Langacker, 2014). Because conceptualizations are never fully shared between interlocutors, human communication is characterized by a fundamental asymmetry between the richness of internal meaning spaces and the limited capacity of external linguistic signals. As a result, meanings cannot, in principle, be fully encoded. While some meanings are explicitly encoded through syntactic structures (Syntax), some are conveyed via pragmatic signals, such as gestures, gaze, and contextual inference, for successful communication (Pragmatics). This reliance on pragmatic inference (Scott-Phillips, 2014) implies that human language is fundamentally underspecified. Understanding how the boundary between syntax and pragmatics arises through cultural evolution is a critical challenge for language science. This study investigates whether linguistic structure can emerge under underspecification, without relying on pragmatic cues during language evolution.

The Iterated Learning Model (ILM), as a foundational framework for the cultural evolution of language, has demonstrated the gradual emergence of systematicity or compositionality under the dual pressures of learnability and expressivity through repeated cycles of parental production and child's learning (Kirby, 2002; Kirby, Tamariz, Cornish, & Smith, 2015). Most ILM-based studies, including the powerful Neural Iterated Learning Model (N-ILM) (Bunyan, Bullock, & Houghton, 2025), have ideally considered language evolution as progress toward a specific one-to-one relationship between signal and meaning under trade-off between compressibility and expressibility. Thus, they have typically assumed a signal-meaning symmetry, where semantic unit matches signal unit. However, this idealized simple condition obviously allows all meanings to be explicitly (compositionally) encoded, effectively eliminating the problem of how the boundary between syntax and pragmatics arises.

Human language, however, is underspecified: the meaning space (M) exceeds the signal space (S). To capture this property, we modify N-ILM (Bunyan et al., 2025) by introducing a persistent meaning–signal asymmetry ($M > S$). In addition, we redefine meaning representations in our model as comprising not only objective information about the external world, but also a subjective component shaped by language users' conceptualization. We operationalize this subjective component by introducing a "conceptualization bit" which encodes binary conceptual distinctions (e.g., active vs. passive construal or figure–ground perception) , implemented as a learner's bias (see Section 2.1 for details). This allows us to test whether compositional structure can emerge under underspecification, where signals cannot fully encode all semantic distinctions.

In our simulation, we aim to clarify how the boundary between syntax and pragmatics emerges during language evolution—specifically, whether underspecified meanings become systematically encoded in syntactic structure or remain to be resolved through pragmatic inference.

2. Methodology

We extend the N-ILM (Bunyan et al., 2025) to simulate language transmission under signal–meaning asymmetry[1]. We implement underspecification by setting the dimensionality of the meaning input layer (M) to exceed that of the signal output layer (S), i.e., $M > S$. One dimension of the meaning vector is reserved for the "conceptualization bit" as described above.

To ensure that our results are not specific to a single operationalization of compositionality, we employ three metrics of compositionality: Compositionality ($Comp$) (Bunyan et al., 2025), Positional Disentanglement ($PosDis$) (Chaabouni, Kharitonov, Bouchacourt, Dupoux, & Baroni, 2020), and Topographic Similarity ($TopSim$) (Brighton & Kirby, 2006). $Comp$ and $PosDis$

[1] Simulation code: https://github.com/ibukiIWAMURA/evolang2026.git

quantify bit-level compositional structure, focusing on meaning-to-signal ($M \rightarrow S$) and signal-to-meaning ($S \rightarrow M$) mapping consistency, respectively, whereas $TopSim$ measures global structural similarity between meaning and signal spaces.

2.1. *Learning Architecture*

Here, language is defined as a mapping between meaning and signal spaces. In our simulation, agents undergo an iterated learning process where language is transmitted from a "parent" (producer) to a "child" (learner) across generations. Learning combines supervised and unsupervised training. Meanings and signals are represented as binary vectors of dimensionality M and S, respectively. The output probability vector is binarized by thresholding probabilities at 0.5. Each mapping component is implemented as a fully connected neural network with a single hidden layer of size N_{node}.

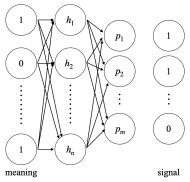

Figure 1. Encoder

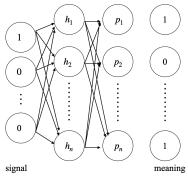

Figure 2. Decoder

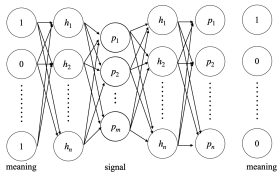

Figure 3. Autoencoder

We train three coupled neural networks with sigmoid activations using stochastic gradient descent (SGD, learning rate 5.0). The meaning space contains 2^M possible meanings, and the training set size is limited to $|B| = 75$, serving as the transmission bottleneck. At each training epoch, we perform the following three steps:

1. The encoder (Fig 1) is trained in a supervised manner on B_1 ($m \rightarrow s$), a set of meaning–signal pairs randomly sampled from the parent's utterances.

2. The decoder (Fig 2) is trained in a supervised manner on B_2 ($s \rightarrow m$), constructed from the same pairs as B_1 but with their order shuffled.

3. The autoencoder (Fig 3), composed of the encoder and the decoder above, is trained in an unsupervised manner on A (m only), a set of 20 meanings sampled independently from B.

In addition, we introduce a conceptualization-bit penalty as a learning bias. During autoencoder training, we apply a weighted loss in which the reconstruction loss for the conceptualization bit (the final meaning bit) is multiplied by $\alpha =$

5. This weighting encourages the model to encode conceptualization distinctions despite the absence of any explicit structural constraints in the meaning vectors. Such binary distinctions are typical in linguistic alternations (e.g., active vs. passive construal or figure–ground perception). Thus, conceptualization is incorporated as a learner's bias rather than as a structural prior.

2.2. *Metrics of Compositionality*

Compositionality refers to the property that the meaning of a whole expression can be derived from the meanings of its parts and the way they are combined (Brighton & Kirby, 2006). Several statistical measures have been proposed to quantify this property. We employ the following three metrics.

- **Compositionality (**$Comp$**)** (Bunyan et al., 2025):

  $Comp$ quantifies bit-level one-to-one correspondence between meaning and signal dimensions, and penalizes cases where multiple meaning bits share the same signal bit. Intuitively, a perfectly compositional language would map each meaning bit to a distinct signal bit without overlap. $Comp$ is defined as

$$Comp = 1 - \frac{1}{n_m} \sum_{i=1}^{n_m} \Big[\min_j H(P(s_j{=}1 \mid m_i{=}1)) \cdot \mathbf{1}_{\text{unique}(i)} + 1.0 \cdot \mathbf{1}_{\neg\text{unique}(i)} \Big]$$

  Here, n_m is the number of meaning bits; i indexes meaning bits; j indexes signal bits; $H(\cdot)$ is the binary Shannon entropy; and $P(s_j{=}1 \mid m_i{=}1)$ is the conditional probability. The indicator functions $\mathbf{1}_{\text{unique}(i)}$ and $\mathbf{1}_{\neg\text{unique}(i)}$ implement the overlap penalty: if multiple meaning bits select the same signal bit, only the one with the lowest entropy (i.e. the smallest $\min_j$) retains its value, while the others receive the maximal penalty of 1.0.

- **Topographic Similarity (**$TopSim$**)** (Brighton & Kirby, 2006):

  $TopSim$ measures the extent to which similarity relations are preserved between meaning and signal spaces, based on the intuition that similar meanings should correspond to similar signals in a compositional language. It is defined as

$$TopSim = r(D_M, D_S)$$

  where r is the Pearson correlation coefficient, D_M and D_S are the pairwise distance matrices over meanings (Hamming distance) and signals (edit distance), respectively.

- **Positional Disentanglement** ($PosDis$) (Chaabouni et al., 2020):

  $PosDis$ quantifies the degree to which each signal position corresponds to a single meaning dimension. The intuition behind this metric is that, in a compositional language, each signal position encodes unique information about one meaning component. $PosDis$ is defined as

$$PosDis = \frac{1}{C_{len}} \sum_{j=1}^{C_{len}} \frac{I(s_j; a_1^j) - I(s_j; a_2^j)}{H(s_j)}$$

  Here, C_{len} is the signal length; $I(s_j; a)$ denotes mutual information between the j-th signal bit and one meaning bit a; $a_1^j = \arg\max_a I(s_j; a)$ is the meaning bit with the highest mutual information; $a_2^j = \arg\max_{a \neq a_1^j} I(s_j; a)$ is the meaning bit with the second highest mutual information; and $H(\cdot)$ denotes the entropy.

For all metrics, higher values indicate greater compositionality. Note that $Comp$ assumes one-to-one correspondence between the meaning and signal dimensions, and is therefore not directly applicable under the asymmetric condition ($M > S$). When $M > S$, multiple meaning bits necessarily share the same signal bit, which confounds the interpretation of $Comp$. We therefore report $Comp$ only for the symmetric condition ($M = S$) to replicate previous results (Bunyan et al., 2025), and focus on $TopSim$ and $PosDis$ for the main analysis under $M > S$.

3. Results

We first report the baseline condition where $M = S = 8$ with no conceptualization-bit penalty ($\alpha = 1$). Figures 4-6 show the results averaged over five independent runs. Thick lines indicate the across-run average, and thin lines indicate individual runs. The x-axes show generations (1-50), and the y-axes show the corresponding metric values (higher = more compositional). All three metrics approach 1.0 under the baseline setting, replicating the results of Bunyan et al. (2025). This suggests that the three metrics behave similarly in our experiments under the symmetric condition ($M = S = 8$).

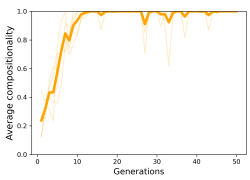

Figure 4. Comp

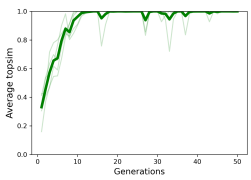

Figure 5. TopSim

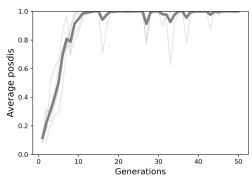

Figure 6. PosDis

We next examine the asymmetric condition with the conceptualization-bit penalty ($M = 8$, $S = 5$, α =5) (Figs. 7- 8). Both $TopSim$ and $PosDis$ remain low throughout transmission [2], suggesting that compositionality does not emerge under this condition. The results for $S = 6$ and $S = 7$ were qualitatively similar.

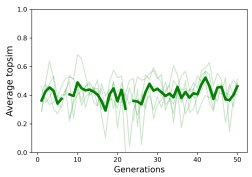

Figure 7. TopSim

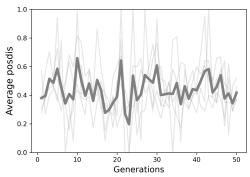

Figure 8. PosDis

4. Discussion

This study investigates whether linguistic structure can emerge under underspecification, without relying on pragmatic cues during language evolution. We modify the Neural Iterated Learning Model (N-ILM) to incorporate a fundamental asymmetry of human language ($M > S$), reflecting the mismatch between rich internal meaning representations and the limited capacity of external linguistic signals. Our results show that when the signal space cannot fully encode all semantic distinctions ($M > S$), compositional structure does not reliably emerge. This suggests that pragmatic inference may play an important role in overcoming underspecification, and may be necessary for the emergence of linguistic structure in more realistic communicative settings.

Importantly, most previous ILM research assume a symmetric setting ($M = S$), which effectively abstracts away from underspecification. While this is a useful experimental idealization, it overlooks the fundamental importance of conceptualization, through which language users construe the world and attempt to encode subjective meanings into signals [3]. Our results highlight a key modelling implication: under the symmetry assumption, learners are not required to resolve underspecification in the first place. This suggests that the emergence of linguistic structure under realistic underspecification may require mechanisms beyond ver-

[2]For $TopSim$ in Fig. 7, some lines disappear when all signals collapse to a single vector, making all pairwise distances zero and the correlation undefined. These gaps indicate degenerate cases where TopSim cannot be computed.

[3]We thank an anonymous reviewer for pointing out that some ILM studies have explicitly adopted signal–meaning symmetry as an idealization and discussed the trade-off between compressibility and expressivity (Smith, Tamariz, & Kirby, 2013; Kirby et al., 2015)

tical transmission alone, potentially including pragmatic inference as emphasized in Ostensive Inferential Communication (Scott-Phillips, 2014).

At first glance, our findings may appear paradoxical: the learning bias (α) to prioritize the speaker's subjective meanings under underspecification inhibits the emergence of structure in our model, yet human languages are both underspecified and highly structured. One possible resolution is that structure emerges incrementally through communicative interaction. Meanings that are easily and repeatedly inferred may become effectively shared, reducing their underspecification over time. Once such meanings become consistently and reliably inferable, they can be progressively incorporated into explicit linguistic structure. Under this view, linguistic structure does not emerge directly from underspecification, but from the gradual accumulation of meanings stabilized through successful communication. This perspective highlights the role of pragmatic interaction in shaping the evolving boundary between what must be encoded in syntax and what can be left to pragmatics.

5. Conclusion

This study suggests that linguistic underspecification (dimensional asymmetry between meaning and signal spaces; $M > S$) inhibits the emergence of linguistic structure in cultural evolution. Our findings further suggest that pragmatic inference may be required to shape which meanings become structurally encoded.

An important direction for future work is to clarify how underspecification is resolved in communication and incrementally incorporated into structure—particularly how grammatical structures reflecting speaker's conceptualization (e.g. active vs. passive construal) can emerge through cultural evolution.

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